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Escola Superior de Tecnologia

Cycle in **Degree** Course  
2nd Semester of 3rd Year  
Course Unit: **Cloud Computing**  
Lecturer: Osvaldo Santos

## Practical work on VM autoscaling in Azure

### Objectives of this sheet

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Initial contact with the autoscaling technique in Azure Cloud Computing, which allows you to scale out (increase the number of VMs) and scale in (reduce the number of VMs) according to user-defined criteria.

### Bibliography and useful links

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How to autoscale in Azure

<https://learn.microsoft.com/en-us/azure/azure-monitor/autoscale/autoscale-get-started>

Azure docs official website

<https://docs.microsoft.com/en-us/azure/?product=popular>



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## Practical work on VM autoscaling in Azure

In this assignment you will use your ipcbcampus student account to access Azure

**At the end of the job, you must delete all Azure resources so that you can preserve your balance**

### 1) Preparation

- a) All resources will be created in European Regions - for example **UK South**.
- b) Create a resource group called "**class**";
- c) Now let's create a NAT Gateway to give the VMs Internet access, with the following characteristics:
  - i) Name: **my-nat-gateway**
  - ii) Region: **UK South**
  - iii) Resource group: **class**
  - iv) Outbound IP: Create a new public IP address, with the name "**my-public-ip**"
  - v) Subnet: create new, with the name "**my-subnet**"
- d) Create the NAT gateway and wait for it to be successfully created.

### 2) Creating the Virtual Machine Scale Set

- a) In this resource group, create a Virtual Machine Scale Set (Home -> Search -> Virtual Machine Scale Set) called "**my-autoscaling**", with the following characteristics:
  - i) Basics section
    - Resource group: **class**
    - Virtual machine scale set name: **my-autoscaling**
    - Availability zone: **1, 2 and 3** (to spread VMs across different datacentres in the region)
    - Orchestration mode: **uniform**
    - Security Type: **standard**



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- Scaling mode: autoscaling
- Scaling configuration: **configure**
  - Scaling: initial number of VMs: 1, maximum number of VMs: 4
  - Scale Out: threshold 75%, duration 5 minutes, number of instances to add: 1
  - Scale In: threshold 25%, duration 5 minutes, number of instances to reduce: 1
- Ubuntu Server 22.04 LTS image
- Size: Standard\_B1S (1 GB RAM, 1 vCPU)
- Authentication by username/password (instead of SSH public key). Use the `username student` and password `student1234567!`

### ii) Spot section

- Don't change anything in this section

### iii) Discs section

- Standard SSD disc

### iv) Networking section:

- Virtual Network: use the "my-subnet" subnet that was created in the previous point.
- Load balancing options: Azure Load Balancer.
- Select a Load Balancer: create a new balancer called "my-balancer", of the public type, with access via port 80 TCP.

### v) Management Section

- Don't change anything in this section

### vi) Health section

- Don't change anything in this section

### vii) Advanced section



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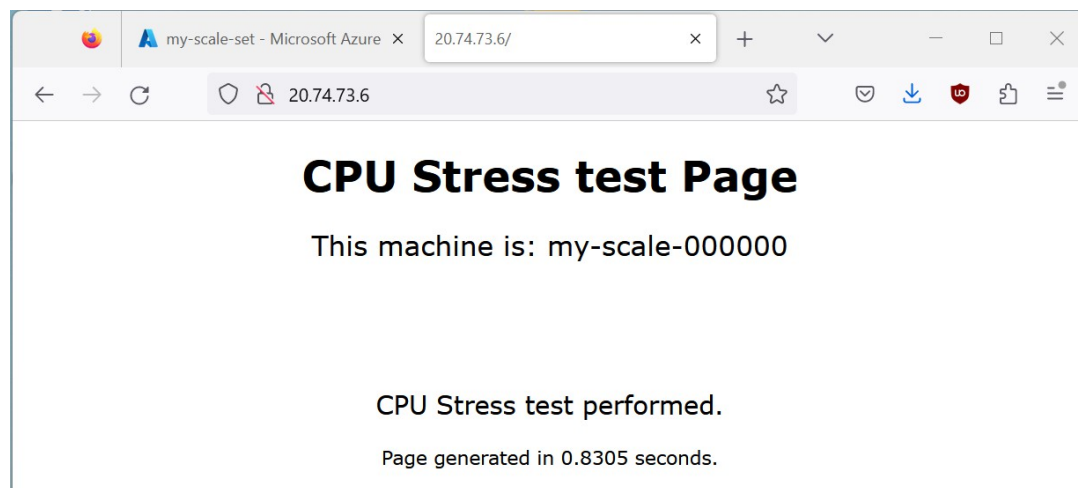


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- Custom data: place here the cloud-init provided by the teacher, which will automatically install Apache, PHP and a PHP script for stress tests.
- b) Now create the Scale Set. Wait for the Scale Set to be created (it may take a few minutes)
- c) Test the web server using a browser and the **Scale Set's** public IP address (don't mistakenly use the IP address of one of the VM instances). You should see a page showing the name of the instance and the time it took for the page to be generated, something like this:





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### 3) Autoscaling tests

- a) In the initial state of Scale Set, how many VM instances are activated? Is the name of the instance that appears in the list the same as the name that appears on the web page?

my-scale-set | Instances

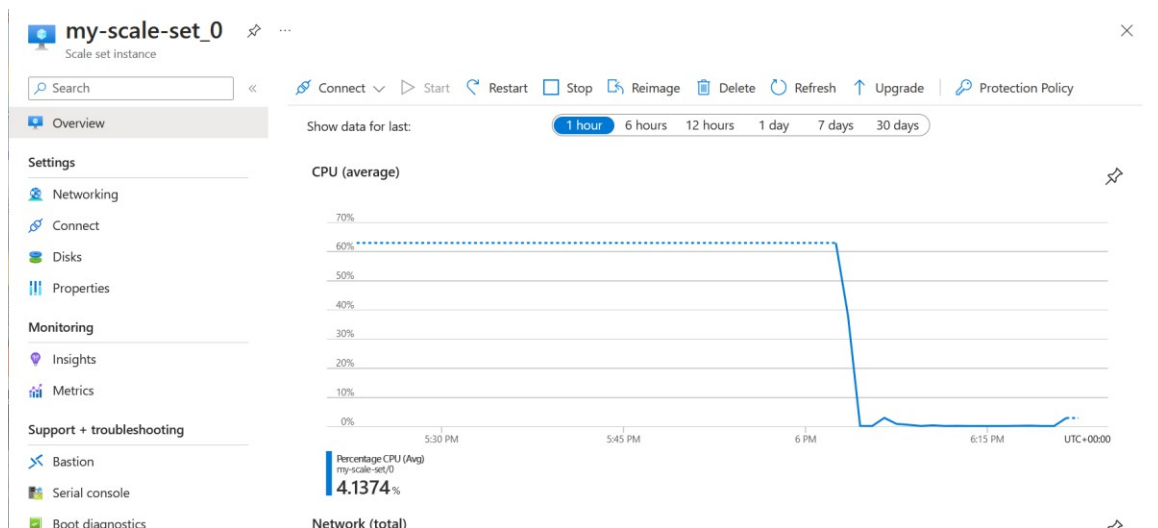
Virtual machine scale set

Search virtual machine instances

| Instance       | Computer name   | Status  | Protection policy | Provisioning sta... | Health state | Latest model |
|----------------|-----------------|---------|-------------------|---------------------|--------------|--------------|
| my-scale-set_0 | my-scale-000000 | Running |                   | Succeeded           |              | Yes          |

Settings: **Instances**, Networking, Scaling, Disks, Operating system

- b) What is the CPU occupancy of the active instance?



- c) Now stress the page using the Apache Benchmark tool (ab). Stress it for at least 30 minutes. Note: you should run the stress test with the IP address of the scale set and not with the IP address of the instance. If the stress tool stops, immediately run it again without waiting too long.

On Windows, this tool (ab.exe) can be downloaded from the Apache binary distribution



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The command has this syntax: `ab -n 100000 -s 250 -c 20 http://40.89.189.105/`

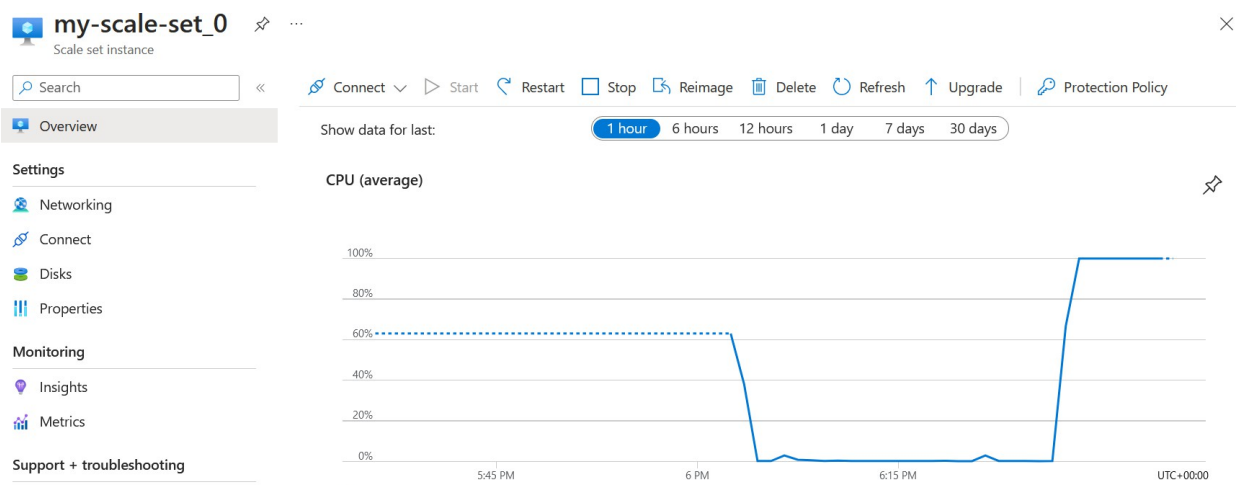
`-n 10000` means that a total of 10000 requests will be made

`-s 250` means that the maximum timeout for each request is 250 seconds

`-c 20` means that 20 open sessions will be held simultaneously

The last parameter is the URL to test (it must have the name of the script or the slash)

- d) After 5 minutes of stress, check the CPU occupancy of the first VM instance. You should see something like this:





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e) After 10 minutes, check how many instances have been created. You should see something like this:

Home > my-scale-set

**my-scale-set** | Instances ☆ ...  
Virtual machine scale set

Search << Start Restart Stop Reimage Delete Upgrade Refresh Protection Policy

Overview  
Activity log  
Access control (IAM)  
Tags  
Diagnose and solve problems

Settings  
Instances

Search virtual machine instances

| Instance                                | Computer name   | Status  | Protection policy | Provisioning state |
|-----------------------------------------|-----------------|---------|-------------------|--------------------|
| <input type="checkbox"/> my-scale-set_0 | my-scale-000000 | Running |                   | Succeeded          |
| <input type="checkbox"/> my-scale-set_1 | my-scale-000001 | Running |                   | Succeeded          |

f) After 20 minutes, check how many instances have been created. You should see something like this:

**my-scale-set** | Instances ☆ ...  
Virtual machine scale set

Search << Start Restart Stop Reimage Delete Upgrade Refresh Protection Policy

Overview  
Activity log  
Access control (IAM)  
Tags  
Diagnose and solve problems

Settings  
Instances  
Networking

Search virtual machine instances

| Instance                                | Computer name   | Status  | Protection policy | Provisioning sta...↑ |
|-----------------------------------------|-----------------|---------|-------------------|----------------------|
| <input type="checkbox"/> my-scale-set_0 | my-scale-000000 | Running |                   | Succeeded            |
| <input type="checkbox"/> my-scale-set_1 | my-scale-000001 | Running |                   | Succeeded            |
| <input type="checkbox"/> my-scale-set_2 | my-scale-000002 | Running |                   | Succeeded            |
| <input type="checkbox"/> my-scale-set_3 | my-scale-000003 | Running |                   | Succeeded            |



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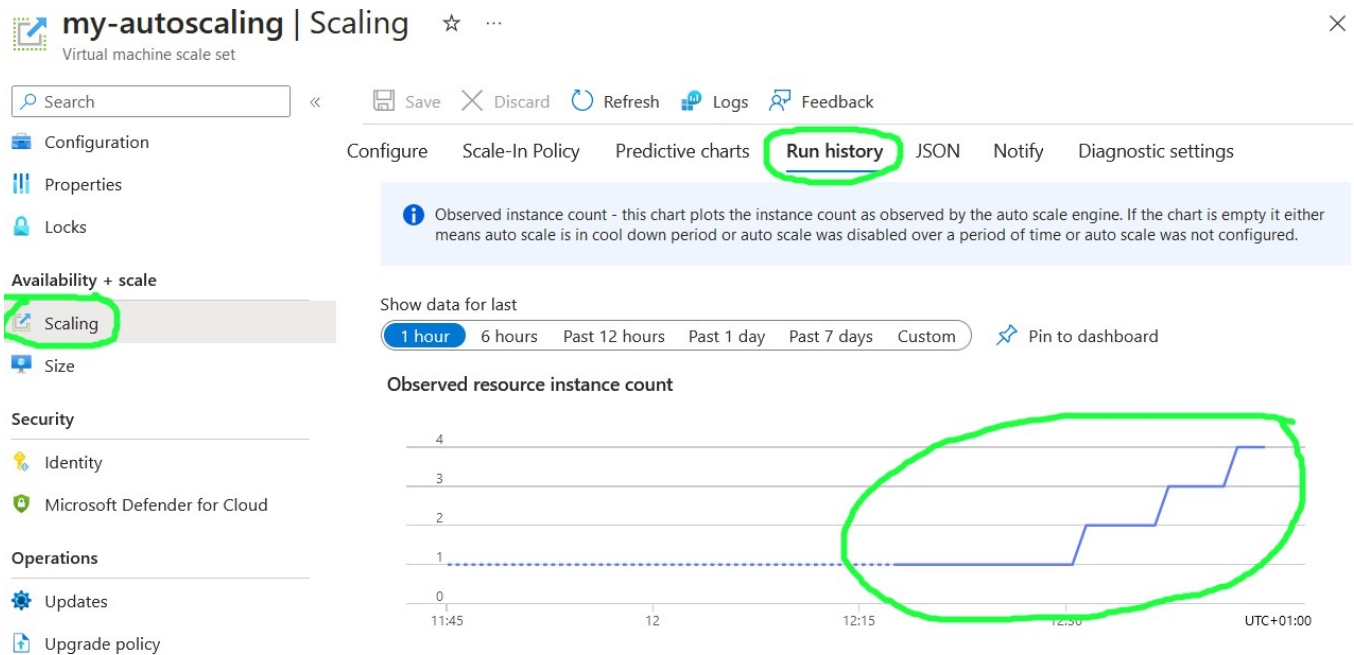


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- g) In the Scaling -> Run history section you can see the evolution of the number of instances over time, something like this:



- h) Now stop the stress test.
- i) After 10 minutes check how many instances are running
- j) After 20 minutes check how many instances are running

#### 4) Conclusion

- a) What are the advantages of this auto-scaling technique?

**Now delete all the resources**